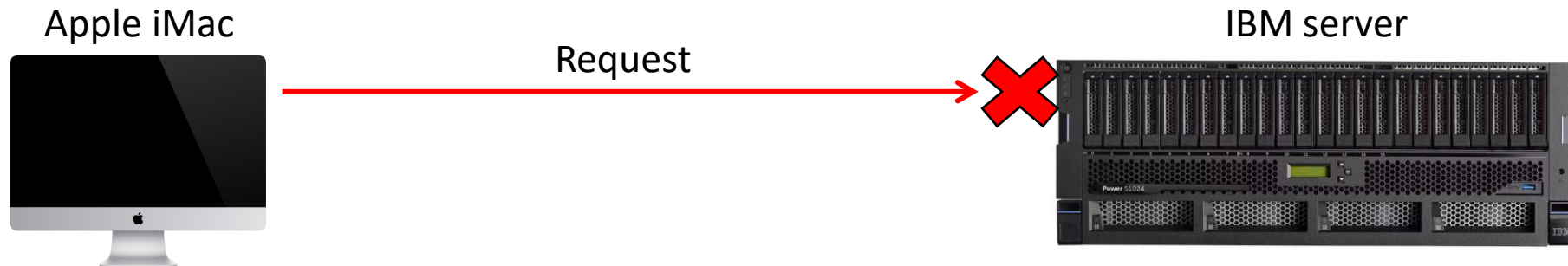


Protocols and Standards

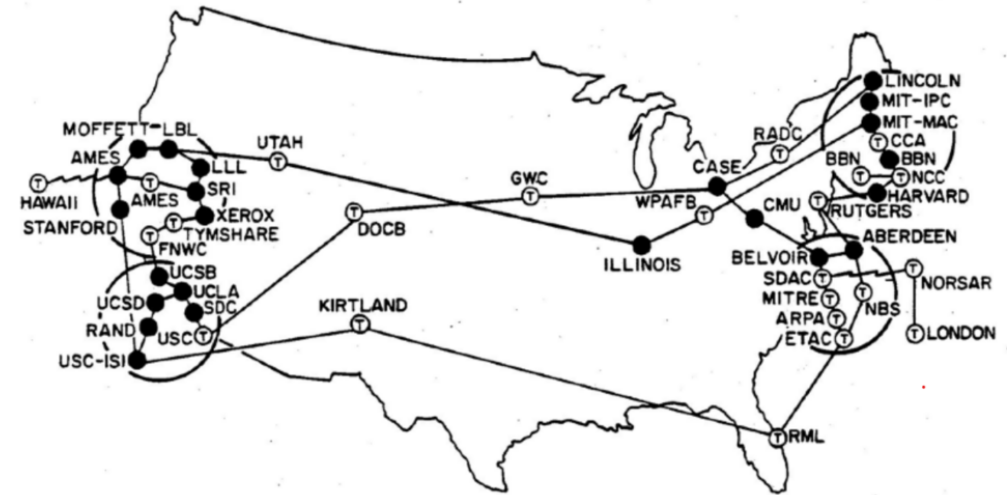
- A **protocol** is a set of rules defining how data should be communicated between devices over a network.
 - The “languages” that computers use to communicate.
- Since the early days of computer networking, there have been several attempts to define the functions needed for computers to communicate with each other.
 - Often developed by a specific vendor (e.g., IBM) to be used with their own products.
 - With a proprietary approach, enabling communications between different vendors’ products was difficult.



- A **standard** is an agreed-upon specification that describes how a protocol or technology should work.
 - With **vendor-neutral** standards, devices of all types can communicate with each other.
 - An Apple MacBook can access a website hosted on a web server running Linux.
 - A PC running Windows can send an email that can be read on a smartphone running Android.

A bit of history

- Early work on the computer networks that would evolve into today's Internet began in the 1960s.
 - The US Department of Defense's **ARPA** (Advanced Research Projects Agency) funded **ARPANET**, which came online in 1969 to connect mainframes at universities and labs.
 - Originally used a protocol called NCP (Network Control Program).
- Vint Cerf and Bob Kahn (working at DARPA) began developing **TCP** (Transmission Control Program) in 1974.
 - Later divided into two protocols still used today:
 - **Transmission Control Protocol (TCP)**
 - **Internet Protocol (IP)**
- These two protocols form the foundation of the protocol suite known as TCP/IP today.
 - ARPANET fully switched to TCP/IP on January 1, 1983.
- TCP/IP became dominant over vendor-proprietary solutions at the time because it was published as a set of open standards that any vendor could implement, and it could run over many different types of networks.



"Arpanet in the 1970s" – Semaforo GMS, CC BY-SA 4.0, via Wikimedia Commons.

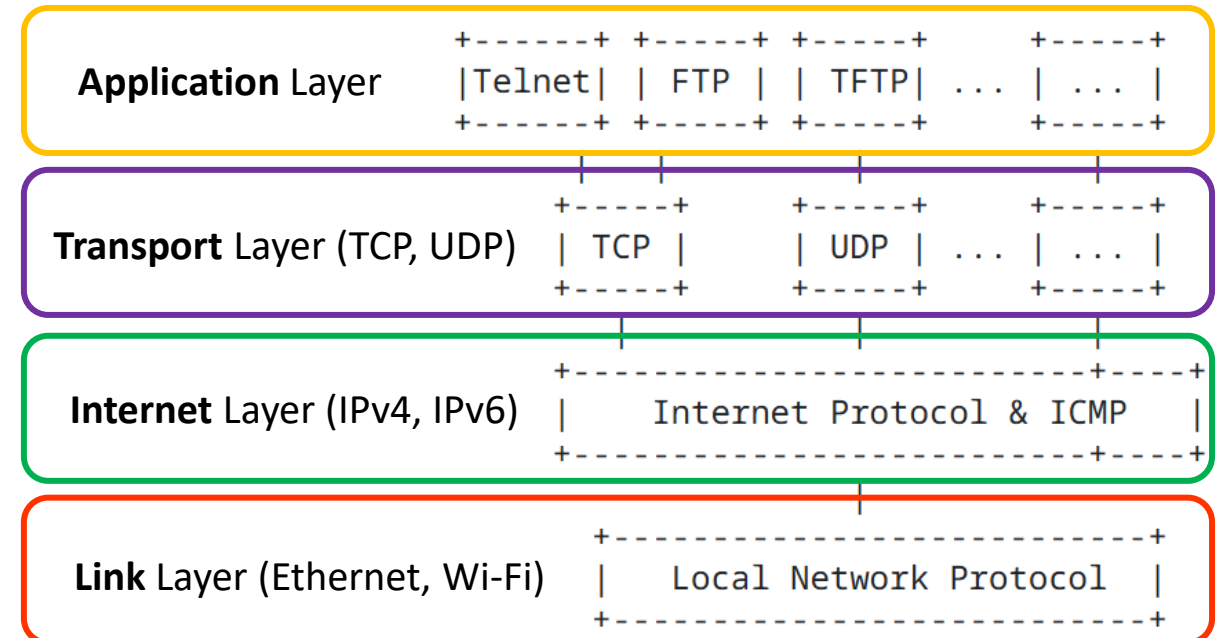
Who defines the standards?

- Most networking standards are developed by independent standards organizations, not by a single vendor, with participation from engineers at many companies.
- **IEEE (Institute of Electrical and Electronics Engineers)**
 - Develops many of the technologies used on local area networks:
 - **Ethernet (802.3)**
 - **Wi-Fi (802.11)**
- **IETF (Internet Engineering Task Force)**
 - Open community that defines protocols used on the Internet:
 - TCP, IP, UDP, HTTP, DNS, etc.
 - Publishes standards in documents called **RFCs** (Requests for Comments).



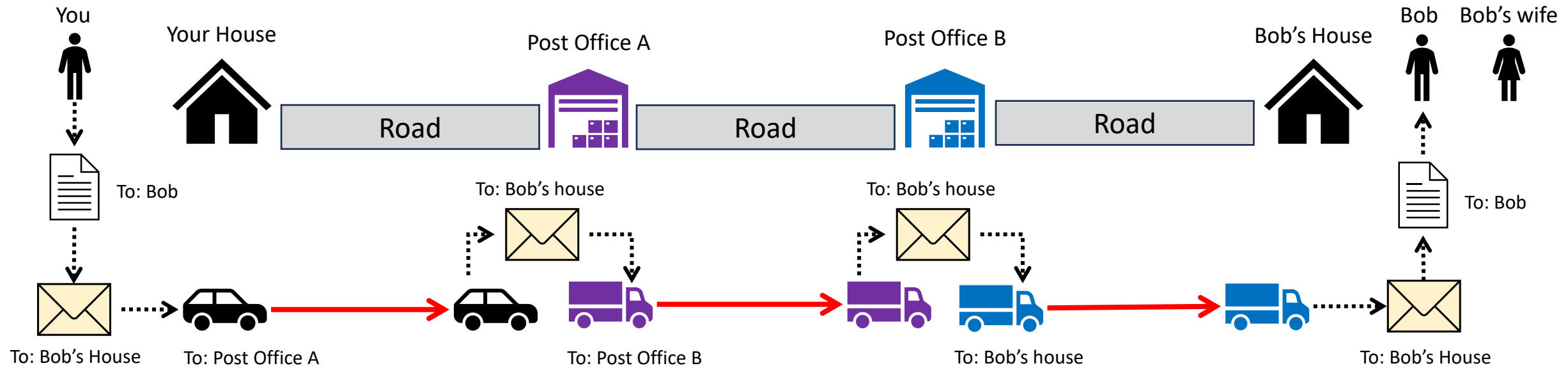
Layered models

- Networks do a lot of different jobs to move data from one computer to another.
 - Physical transmission of signals, local delivery on a LAN, routing traffic between networks, end-to-end conversations, applications, etc.
- A model lets us group related jobs into layers.
 - Each layer has a specific role.
 - Each layer uses the services of the layer below and provides services to the layer above.
- Protocols live (mostly) at one layer.
 - Examples later: IP, TCP, HTTP, etc.
 - Together they form a **stack** of protocols that work as a team (the **network stack**).
- The model is a description, not a law.
 - Different textbooks/courses use slightly different models (4-layer, 5-layer, etc.).



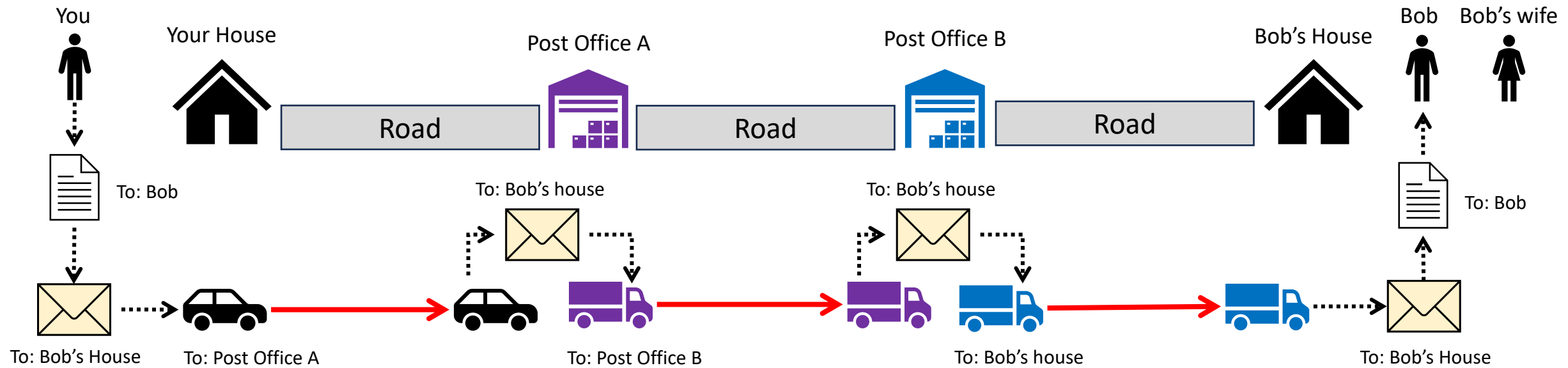
From RFC 791, "Internet Protocol"
(1981)

Sending a letter



- You write a letter addressed to your friend, **Bob**.
- You put the letter in an envelope addressed to **Bob's house**.
- You deliver the envelope in your car to **post office A**.
 - Post office A moves the envelope to a truck and delivers it to **post office B**.
 - Post office B moves the envelope to a new truck and delivers it to **Bob's house**.
- The letter, addressed to **Bob**, is read by Bob.

Building a model



Content layer: the text of the letter, what you actually want to say.

Recipient layer: "To: Bob" vs "To: Bob's wife": the intended recipient inside the house.

Address layer: the intended destination address for the house where the recipient lives.

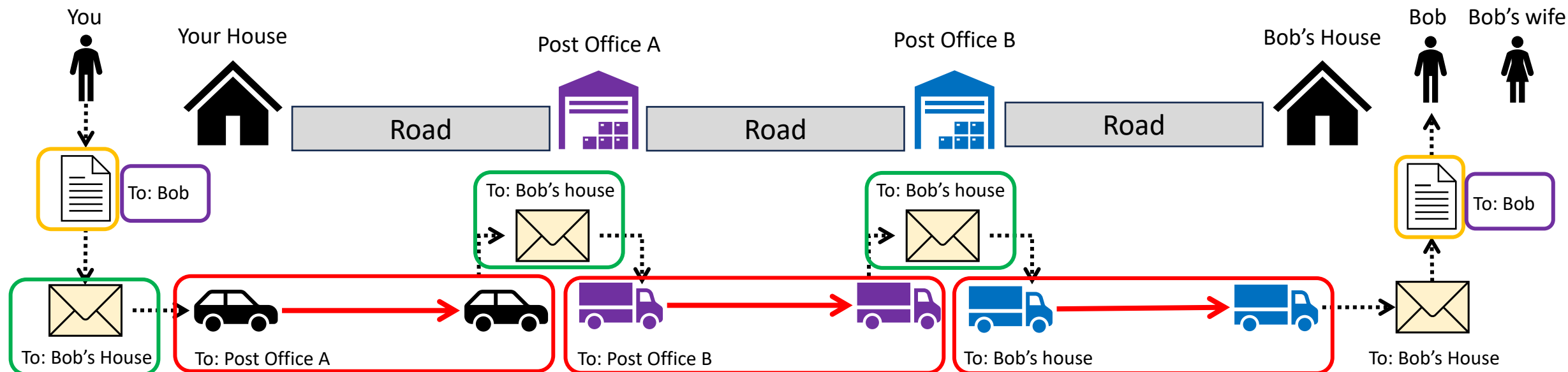
Local Delivery layer: delivery to the next stop on the path using cars/trucks: post office A, then post office B, then Bob's house.

Infrastructure layer: the roads that carry the cars and trucks.

Related layers:

- Travel by ground?
→ use a car or truck
- Travel by air?
→ use an airplane
- Travel by water?
→ use a ship

Separation of layers



Content layer

Recipient layer

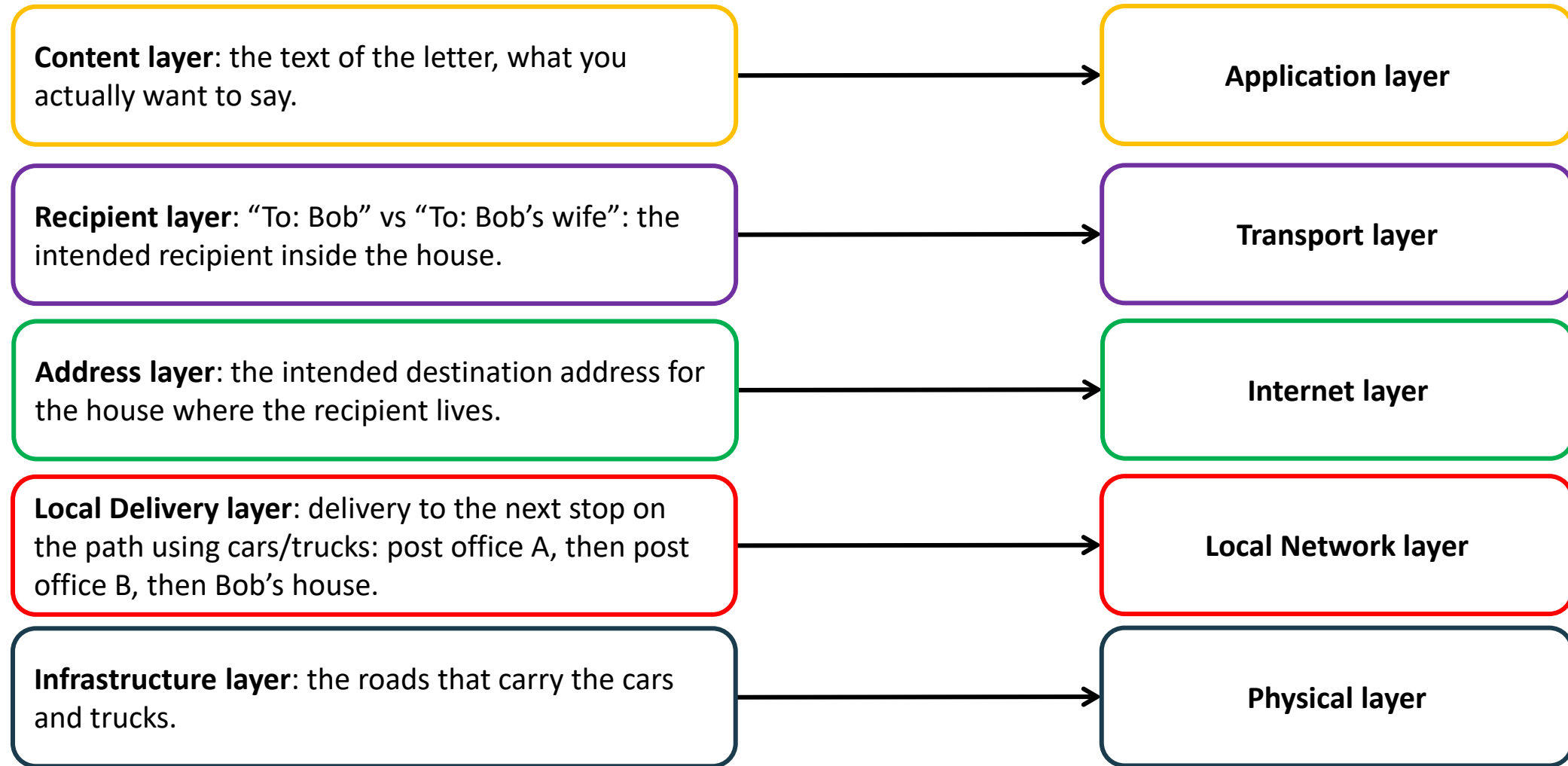
Address layer

Local Delivery layer

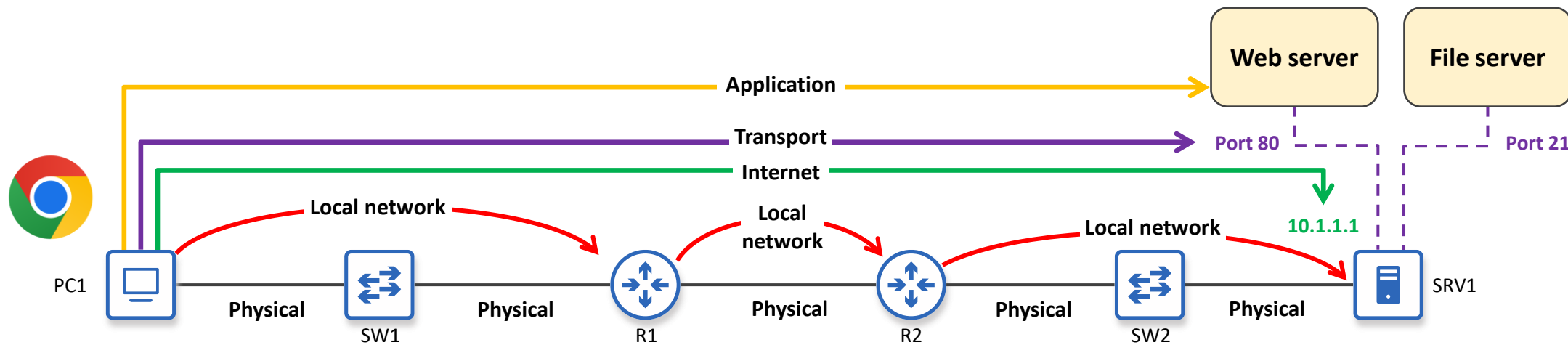
Infrastructure layer

- Each layer has its own job.
 - Layers work together to deliver the message, but each one focuses on its own task.
- What happens inside one layer doesn't change the job of the other layers.
 - Changing the content of the letter doesn't change the delivery steps.
 - Changing the delivery path doesn't affect the letter itself.

The TCP/IP model



The TCP/IP model



Application layer

- Protocols for communication between application processes; create and interpret the data.

Transport layer

- Provides end-to-end communication between application processes using **port numbers**.

Internet layer

- Provides end-to-end communication between hosts across networks using **IP addresses** and routers.

Local Network layer

- Provides hop-to-hop delivery within a local network using **MAC addresses** and switches.

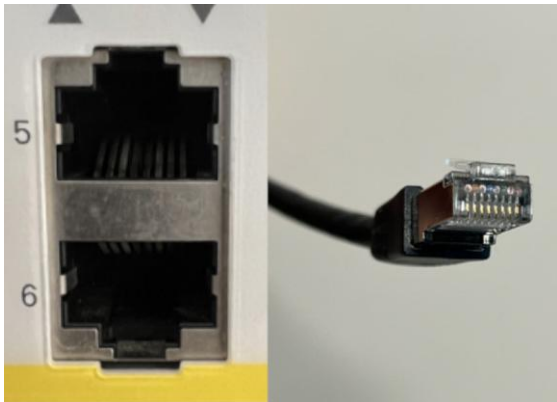
Physical layer

- Sends bits as electrical, optical, or radio signals over the physical medium.

Layer 1: The Physical layer



- The **Physical Layer** (Layer 1) sends and receives bits as electrical, optical, or radio signals over the medium.
- Defines things like cables, connectors, signal levels, and link speeds.
- Examples: copper UTP cables, fiber-optic cables, Wi-Fi radios and antennas, network interface cards (NICs).
- The physical aspects of transmitting data are very complex.
 - Network engineers typically don't have to know the low-level details.



By Barcex - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=10391537>

Application layer

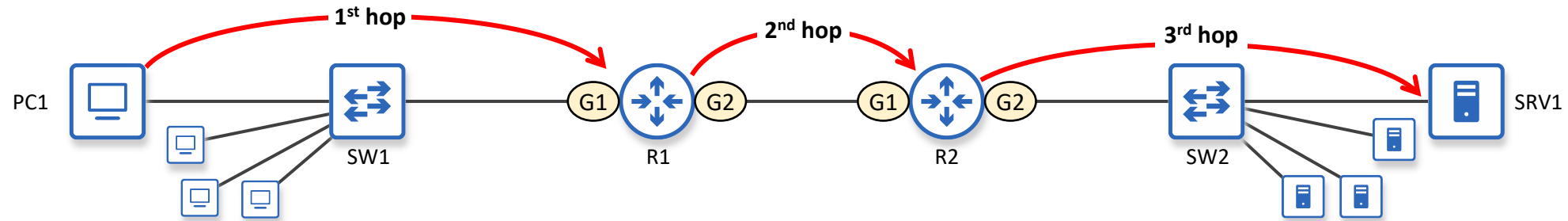
Transport layer

Internet layer

Local Network layer

Physical layer

Layer 2: The Local Network layer



- The **Local Network layer** (Layer 2) provides **hop-to-hop** delivery of messages on a local network.
 - A **hop** is one step along the path between two devices:
 - from one router or host, to the next router or host in the path.
 - Switches don't count: a switch just extends the local network, allowing multiple devices to connect.
- Uses **MAC (Media Access Control) addresses** to identify interfaces.
 - **PC1** sends the message to the MAC address of **R1's G1 interface (NIC)**.
 - **R1** sends the message to the MAC address of **R2's G1 interface (NIC)**.
 - **R2** sends the message to the MAC address of **SRV1's interface (NIC)**.
- Protocols at this layer include:
 - Ethernet (IEEE 802.3)
 - Wi-Fi (IEEE 802.11)

Application layer

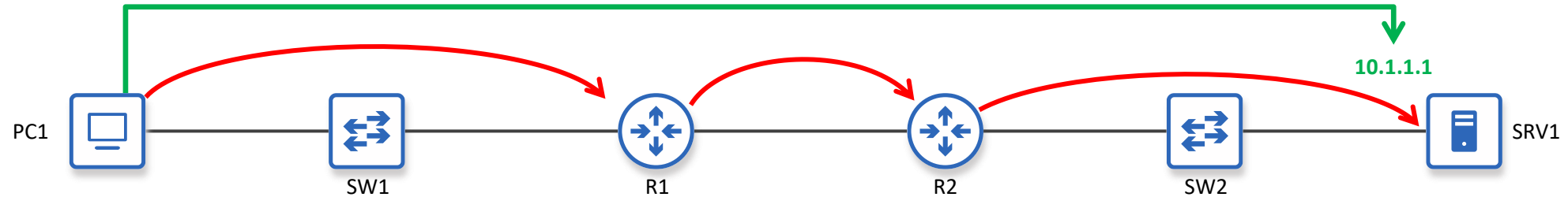
Transport layer

Internet layer

Local Network layer

Physical layer

Layer 3: The Internet layer



- The **Internet layer** (Layer 3) provides end-to-end delivery between hosts across multiple networks.
 - Internet = internetwork (between networks)
- Uses **IP addresses** to identify hosts in the network.
 - When PC1 sends a message to SRV1, it addresses the message to SRV1's IP address.
- **Routers** operate mainly at this layer, using the message's destination IP address to forward the message toward its final destination host.
- Protocols at this layer include:
 - IP (IPv4, IPv6)
 - ICMP (Internet Control Message Protocol)

Application layer

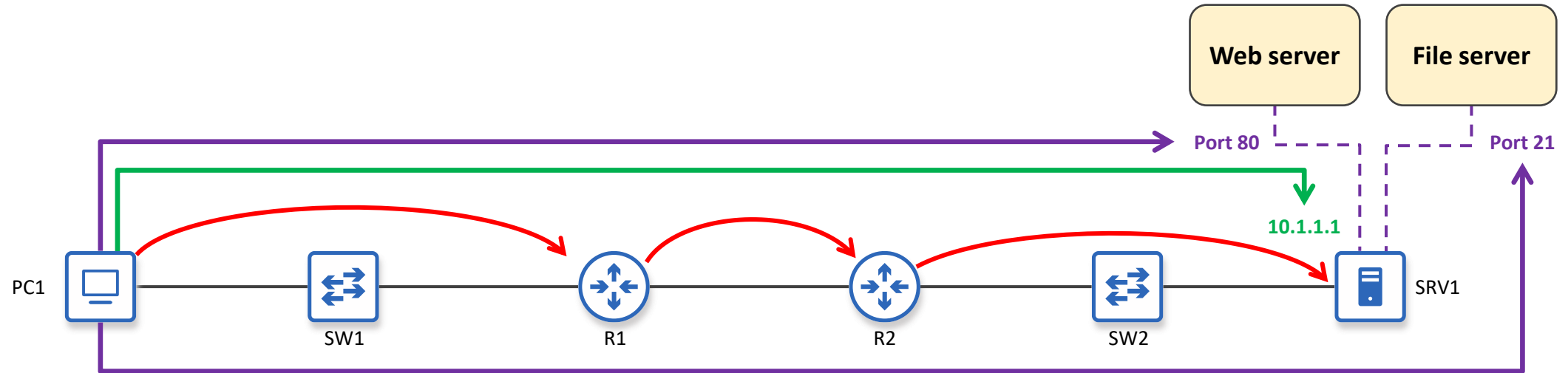
Transport layer

Internet layer

Local Network layer

Physical layer

Layer 4: The Transport layer



- The **Transport layer** (Layer 4) provides end-to-end communication between application processes.
 - Also called “process-to-process” or “service-to-service”.
- Uses **port numbers** to identify the processes on each host.
 - When the web client on PC1 wants to send a request to the web server running on SRV1, it addresses the message to port 80.
- Runs mainly on the communicating hosts (PC1 and SRV1); routers normally operate based on IP (Layer 3), not on Transport-layer information.
- Protocols at this layer include:
 - UDP (User Datagram Protocol): simple and efficient
 - TCP (Transmission Control Protocol): more robust features beyond basic message addressing

Application layer

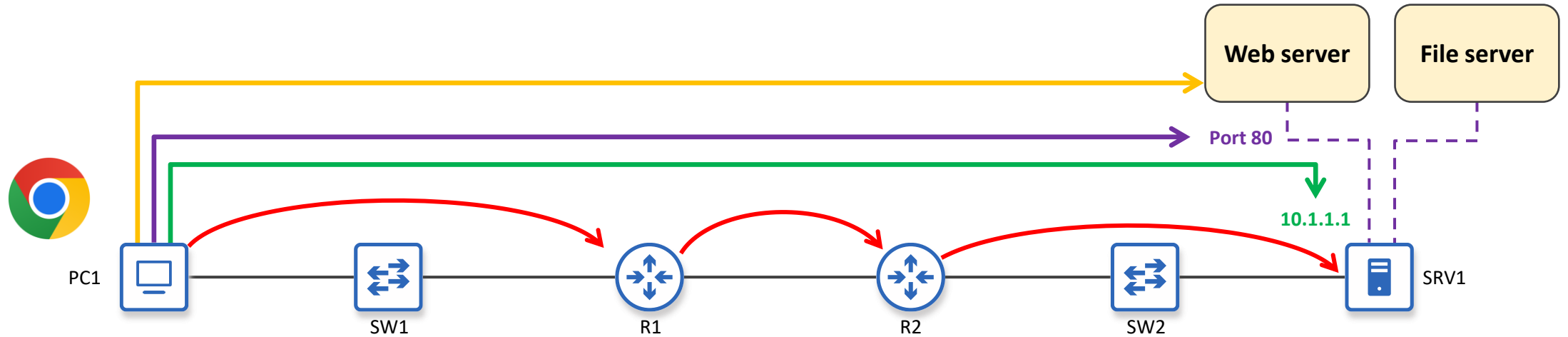
Transport layer

Internet layer

Local Network layer

Physical layer

Layer 5: The Application layer



- The **Application layer** (Layer 5) is where network communications meet applications.
 - Usually called **Layer 7** (more info on that later...)
- Defines how application processes format, send, and interpret data.
- Protocols at this layer define message formats and rules for specific tasks, such as:
 - Browsing web pages (HTTP/HTTPS)
 - Transferring files (FTP, TFTP)
 - Sending/receiving email (SMTP, POP3, IMAP)
- Network infrastructure devices (routers, switches) don't care about Application-layer details.
 - They just move messages across the network.
 - Only the communicating hosts interpret the data.

Application layer

Transport layer

Internet layer

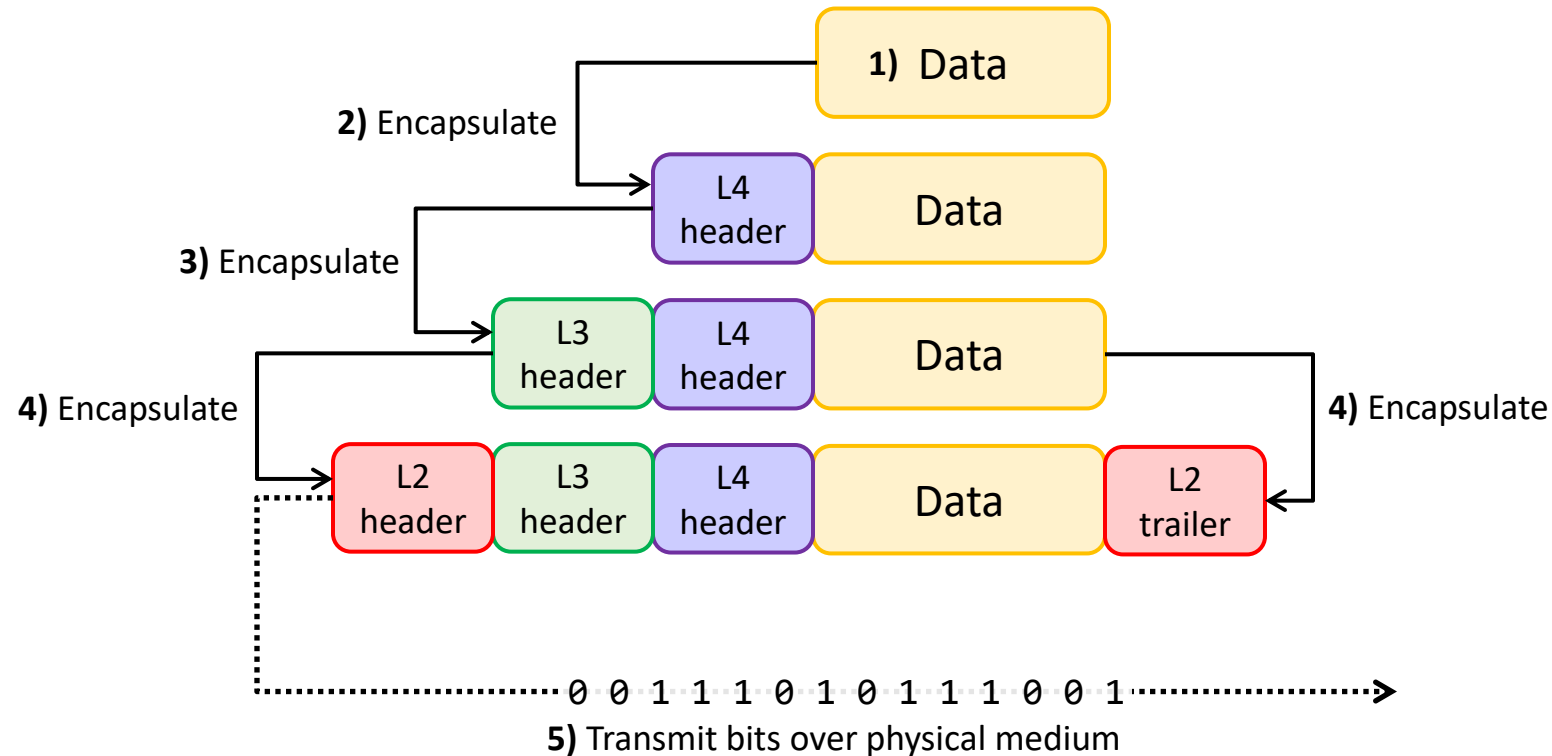
Local Network layer

Physical layer

Encapsulation & decapsulation

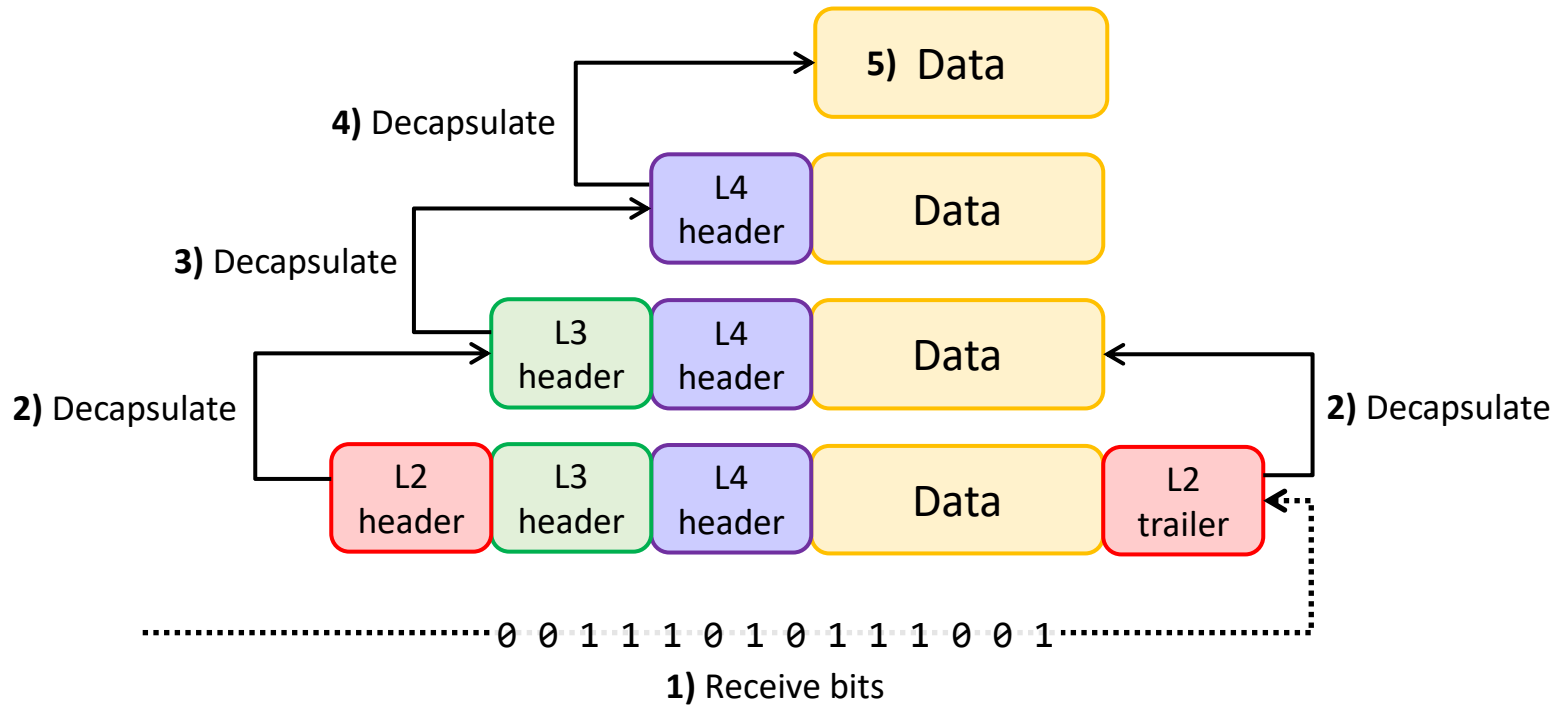


5	Application
4	Transport
3	Internet
2	Local Network
1	Physical



- The **Application layer** prepares the data to be sent over the network.
- As the message moves down the stack, each layer **encapsulates** the data with a **header** including the information needed for that layer.
 - Source and destination addresses (port numbers, IP addresses, MAC addresses), etc.
 - Layer 2 also adds a **trailer** that the receiving device uses to check for transmission errors.
- The **Physical layer** transmits the bits as signals over the physical medium.
 - The **L2 header** is transmitted first, and the **L2 trailer** is transmitted last.

Encapsulation & decapsulation



Application	5
Transport	4
Internet	3
Local Network	2
Physical	1

- The receiving device receives the message as a stream of bits at Layer 1.
- The device examines the information in the Layer 2 header and trailer, and then removes them (**decapsulation**).
 - The decapsulation process continues up the stack: Layer 3 removes the L3 header, then Layer 4 removes the L4 header, and then the data is delivered to the Application layer.
- The application processes the data and, if needed, generates a response that goes back down the stack.

PC1

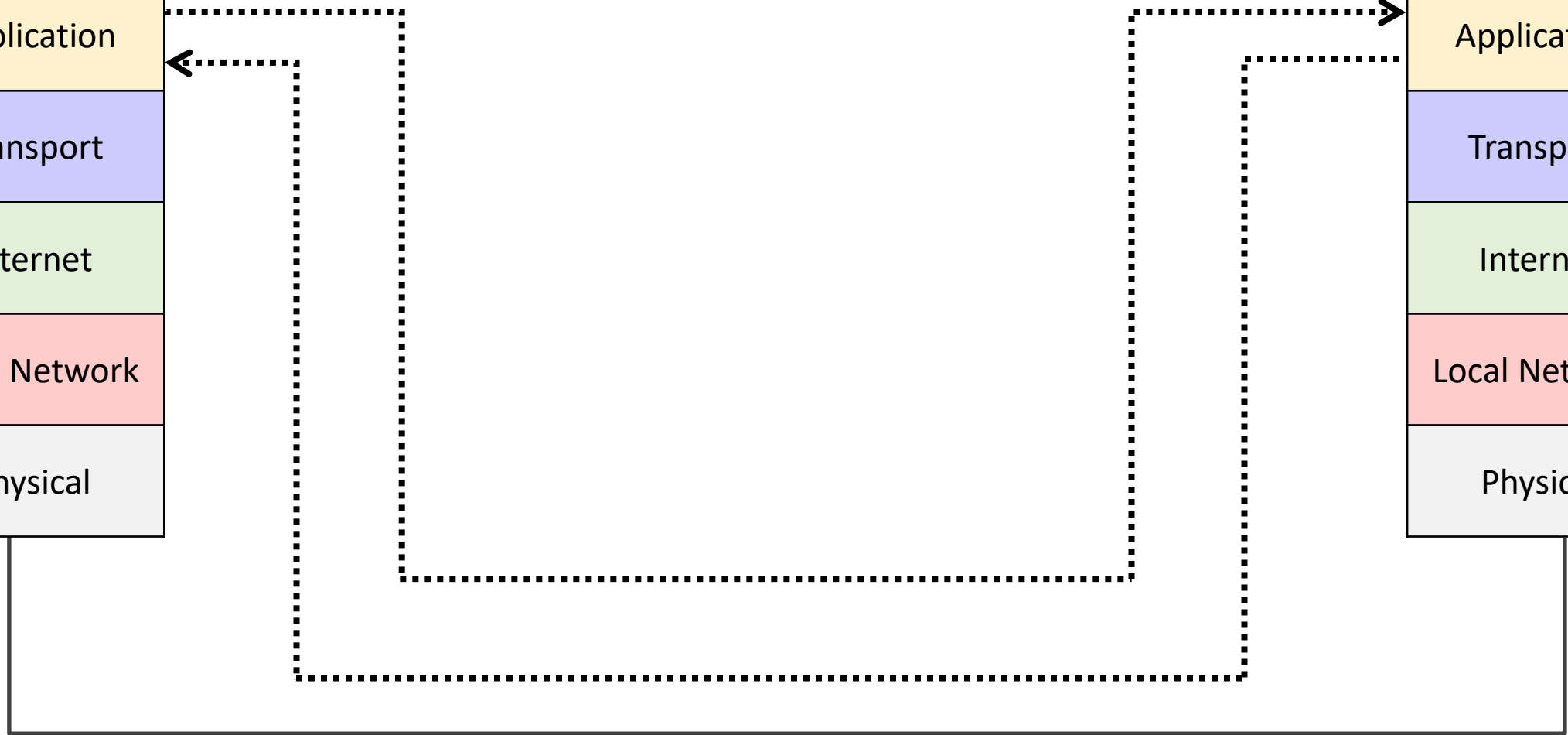


SRV1



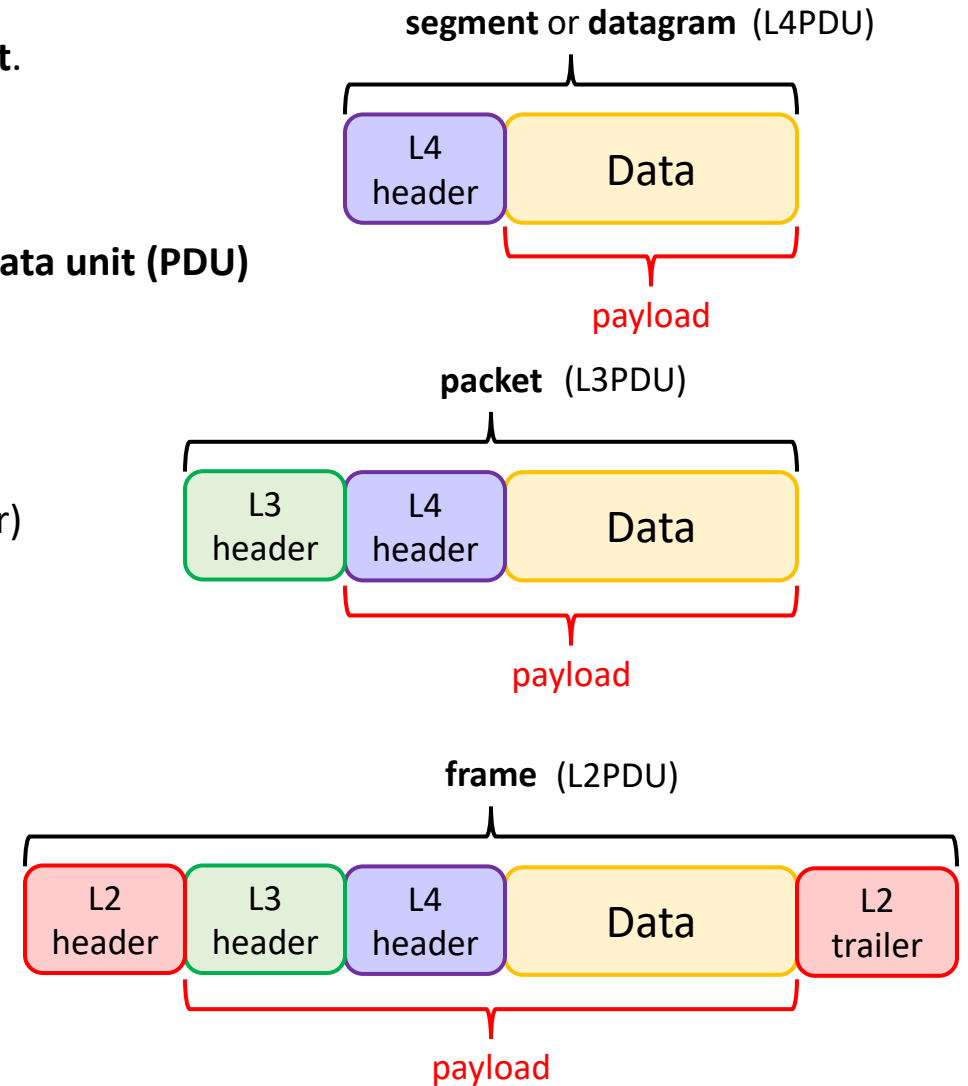
5	Application
4	Transport
3	Internet
2	Local Network
1	Physical

5	Application
4	Transport
3	Internet
2	Local Network
1	Physical



Protocol data units

- At each stage in the encapsulation/decapsulation process, there is a name given to the message:
 - The combination of data and a L4 header is called a **segment** (TCP) or **datagram** (UDP).
 - TCP creates segments, UDP creates datagrams.
 - The combination of a **segment/datagram** and a L3 header is called a **packet**.
 - The combination of a packet and a L2 header/trailer is called a **frame**.
 - This is what is actually sent over the wire.
- We can use alternative names to describe the message at each stage: **protocol data unit (PDU)**
 - A segment or datagram is a **Layer 4 PDU (L4PDU)**.
 - A packet is a **Layer 3 PDU (L3PDU)**.
 - A frame is a **Layer 2 PDU (L2PDU)**.
- The contents of each PDU (everything encapsulated by that layer's header/trailer) are called the **payload**.
 - A **segment or datagram's** payload is the **application data**.
 - A **packet's** payload is a **segment or datagram**.
 - A **frame's** payload is a **packet**.

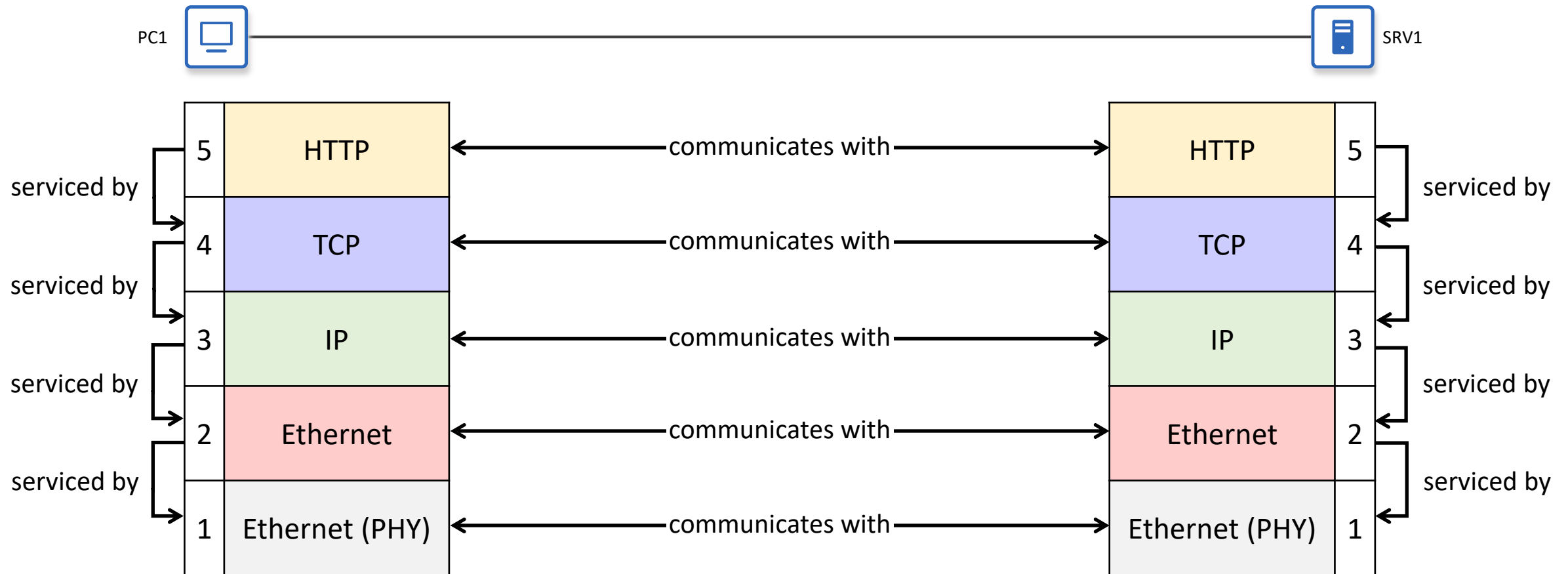


Adjacent-layer interaction



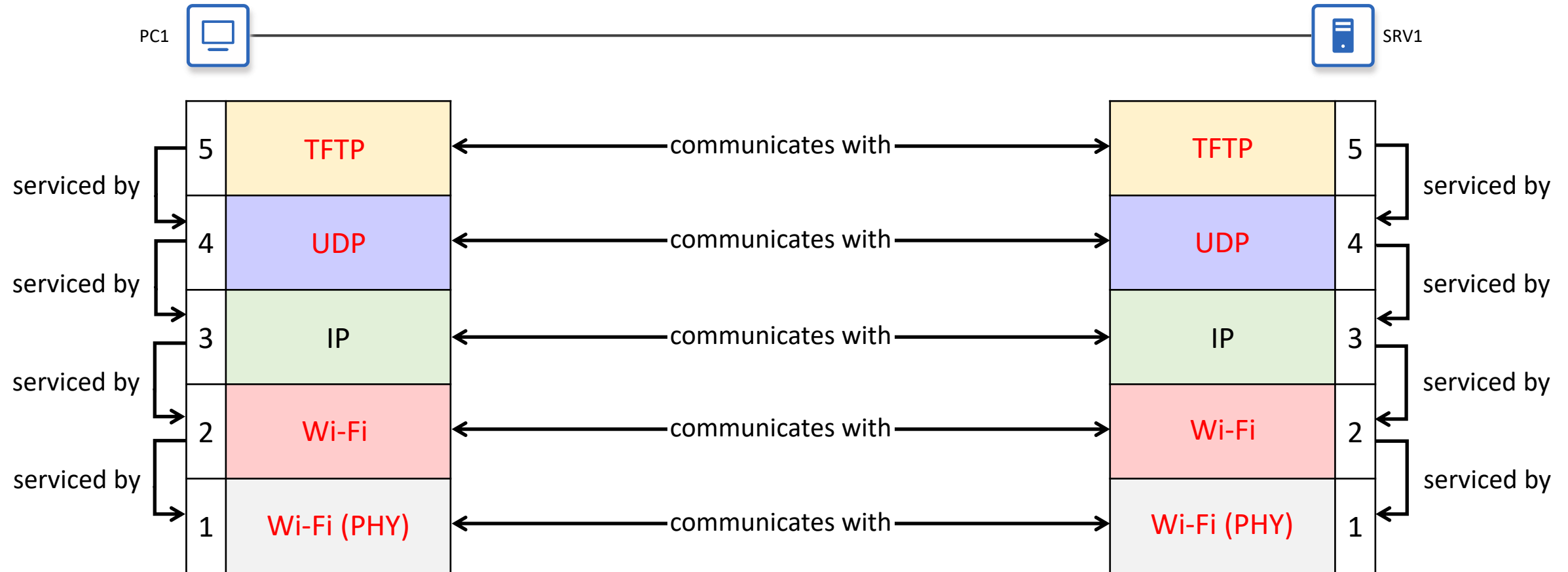
- Each layer provides a service to the layer above it, and is serviced by the layer below it (**adjacent-layer interaction**).
 - Layer 4** provides a service to **Layer 5** by delivering data to the correct application using port numbers.
 - Layer 3** provides a service to **Layer 4** by delivering segments/datagrams to the correct destination host using IP addresses.
 - Layer 2** provides a service to **Layer 3** by delivering packets to the next hop using MAC addresses.
 - Layer 1** provides a service to **Layer 2** by sending and receiving frames as electrical, optical, or radio signals.

Same-layer interaction



- Each layer communicates with the same layer on other devices (**same-layer interaction**).
 - The Application layer on one host sends data to the Application layer on the other host.
 - A segment/datagram is addressed to the **Layer 4 port number** of the correct application on the destination host.
 - A packet is addressed to the **Layer 3 IP address** of the destination host.
 - A frame is addressed to the **Layer 2 MAC address** of the next hop.
 - Signals sent out of a physical port are received by a **physical port** on the connected device.

Separation of layers



- Each layer has its own job and provides a specific service to the layers above.
 - The layers are modular.
- As long as each layer keeps its “contract” with the other layers, we can improve or replace protocols at different layers without redesigning everything.
- That flexibility is one of the main benefits of a layered model.

The OSI model

- TCP/IP development started in the 1970s (ARPANET work, early TCP/IP specs).
- In the late 1970s and 1980s, the **International Organization for Standardization (ISO)** designed a 7-layer **Open Systems Interconnection (OSI) model** and a matching protocol suite.
 - The goal was to create international, vendor-neutral networking standards that could unify existing proprietary stacks and potentially replace TCP/IP.
- Governments, including the US, promoted OSI as the preferred/recommended stack for new deployments.
- OSI protocols ended up being late and complex, and never gained the same deployment as TCP/IP.
 - TCP/IP “won” in the real world, although some OSI technologies are still used.
- Today, almost all real networks use TCP/IP, but the 7-layer OSI model survives as a reference/teaching model and a common way to talk about “layers”.
- Most networking resources use a 5-layer model like the one covered in this video, but with names from the OSI model.
 - That is why the TCP/IP Application layer is often called **Layer 7**.

**Adapted
5-layer model**

7	Application
4	Transport
3	Network
2	Data Link
1	Physical

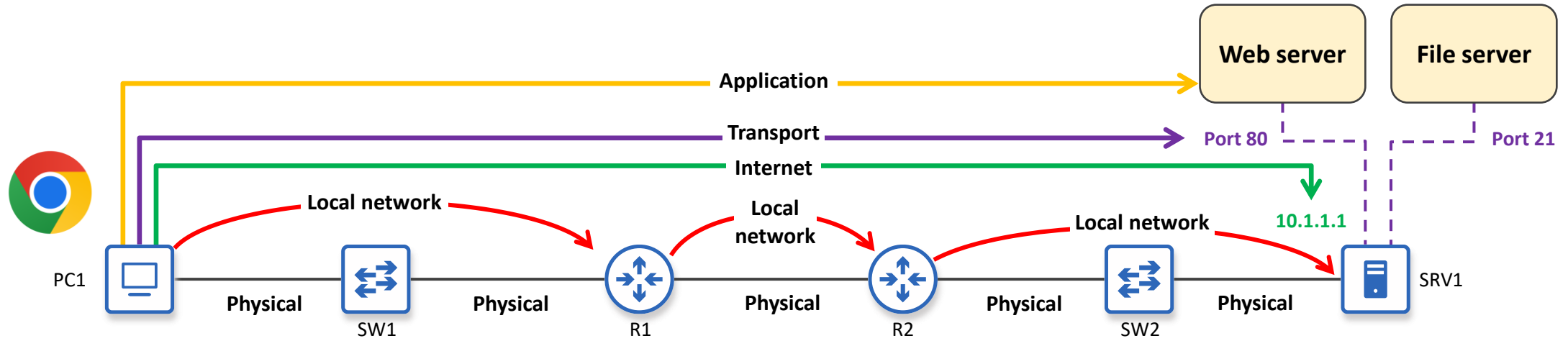
OSI model

7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

Other versions

Arpanet Reference Model (RFC 871)	Internet Standard (RFC 1122)	Internet model (Cisco Academy ^[56])	TCP/IP 5-layer reference model (Kozierok, ^[57] Comer ^[58])	TCP/IP 5-layer reference model (Tanenbaum ^[59])	TCP/IP protocol suite or Five-layer Internet model (Forouzan, ^[60] Kurose ^[61])	TCP/IP model (Stallings ^[62])	OSI model (ISO/IEC 7498-1:1994 ^[63])
Three layers	Four layers	Four layers	Four+one layers	Five layers	Five layers	Five layers	Seven layers
Application/ Process	Application	Application	Application	Application	Application	Application	Application Presentation Session
Host-to-host	Transport	Transport	Transport	Transport	Transport	Host-to-host or transport	Transport
	Internet	Internetwork	Internet	Internet	Network	Internet	Network
Network interface	Link	Network interface	Data link (Network interface)	Data link	Data link	Network access	Data link
—	—	—	(Hardware)	Physical	Physical	Physical	Physical

Review (TCP/IP model)

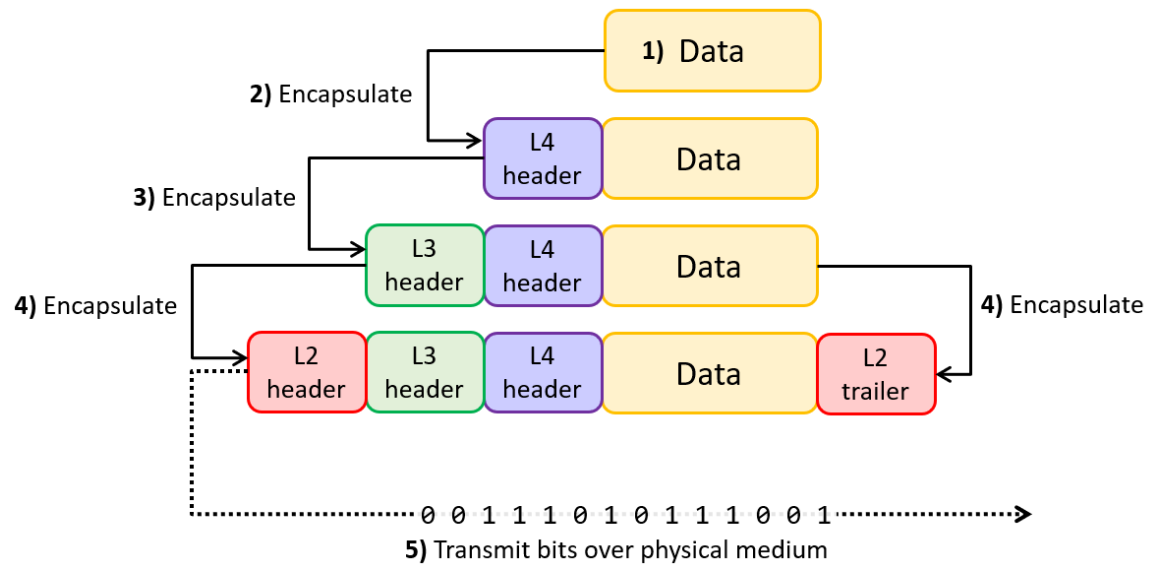


5	Application
4	Transport
3	Internet
2	Local Network
1	Ethernet (PHY)

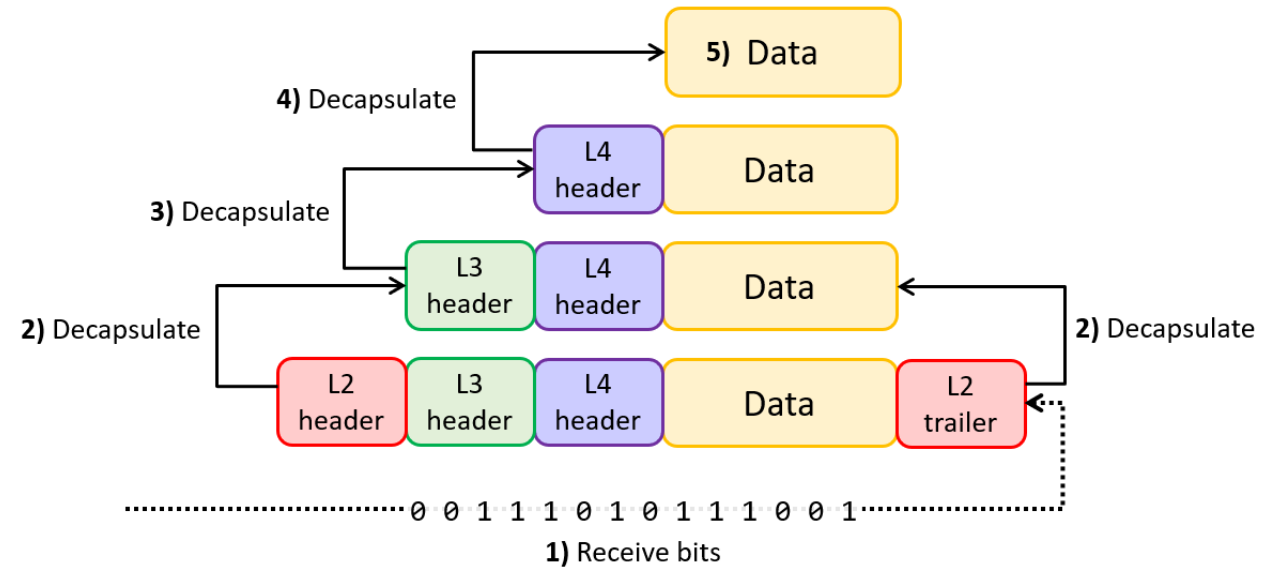
Common 5-layer model

7	Application
4	Transport
3	Network
2	Data Link
1	Physical

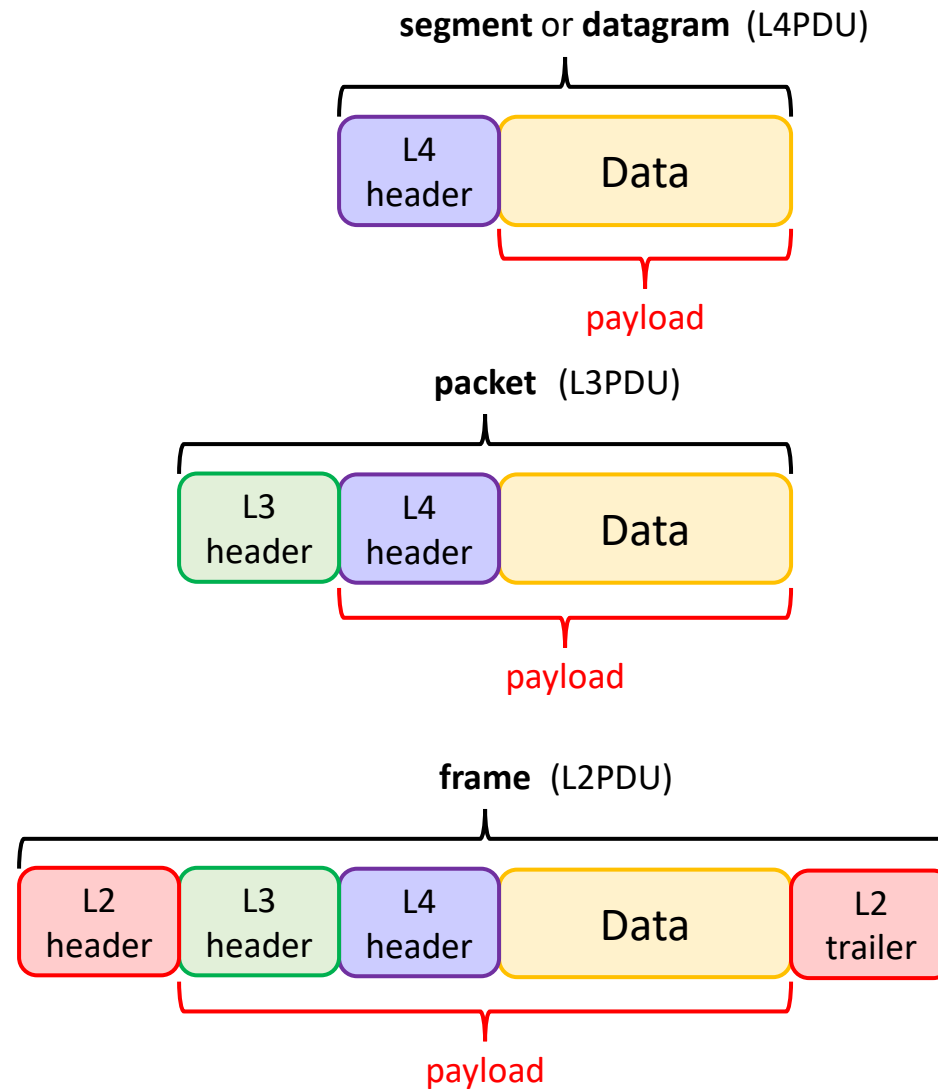
Encapsulation



Decapsulation



Review (PDUs)



Review (Adjacent- and same-layer interaction)

